STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define period.
2. Frequency-period relation?
3. Restoring force in SHM?
4. Where is speed maximum in SHM?
5. Where is acceleration magnitude maximum?
6. Spring angular frequency?
7. Small-angle pendulum period?
8. Total spring energy for amplitude A?

Section B 2-mark questions - 6 x 2 = 12 marks

9. Period .5s: frequency?
10. m=.5kg,k=200N/m: period?
11. For x=Acosomegat, acceleration?
12. Pendulum length becomes 4l: period factor?
13. Velocity at an extreme?
14. Condition for pendulum SHM approximation?

Section C 3-mark questions - 4 x 3 = 12 marks

15. m=.5,k=200: angular frequency and period?
16. A=.1m,omega=4: maximum speed and acceleration?
17. Pendulum length 1m,g=9.8: period?
18. Describe energy in one SHM cycle.

Section D 4-mark questions - 2 x 4 = 8 marks

19. k=50N/m,A=.2m: total energy and U at x=.1m?
20. Why does a real oscillator stop?